survives a careless DELETE typed into phpMyAdmin. The API checks first and returns a helpful message, but the constraint is the thing that guarantees it.

Two values are deliberately not stored. The total cost of a booking is tickets multiplied by the event's ticket_price, and the number of places still available is the event's capacity minus the sum of tickets sold. Both are derived when they are needed. Storing them would create a second copy of the truth that could drift out of step the first time a price changed.

On the events table, latitude and longitude were added as DECIMAL(9,6), which is accurate to about 11 centimetres - far more than a weather grid cell needs, and enough that the same columns could later be used to draw a map. Nullable, because not every venue has been geocoded yet, and the weather panel has to cope with that.

The sample data was extended to 16 registrations spread across five events, comfortably over the required minimum of 10, so that the 'who is going' list on the event page and the registration list on the admin edit screen both have something realistic to show. The whole schema and dataset still export to a single SQL file, database/charityevents-db.sql, as the brief requires.

5. API Design

The API exposes eleven endpoints plus a health check. They are grouped by resource rather than by page, so the shape of the URL says what is being acted on:

Table 3 - The endpoints. Every one of them is called by at least one page.

Method	Path	Purpose	Success	Errors
GET	/api/events	Upcoming events, optionally filtered by date, location, category and free-text search	200	400
GET	/api/events/past	Past events, for the 'previously' section of the home page	200	-
GET	/api/events/:id	One event plus its complete registration list, newest booking first	200	400, 404
POST	/api/events	Create an event (admin)	201	400
PUT	/api/events/:id	Replace an event's details (admin)	200	400, 404
DELETE	/api/events/:id	Delete an event, blocked when bookings exist	200	400, 404, 409
GET	/api/registrations	All bookings, or those for one event via ?event_id=	200	400
POST	/api/registrations	Create a booking (public site)	201	400, 404, 409